

Quiz 10

Illya Starikov

August 7, 2025

Problem #1

$$\begin{array}{ll} & S \\ \implies & AB \quad \text{via } S \rightarrow AB \\ \implies & aXB \quad \text{via } A \rightarrow aX \\ \implies & aXbYd \quad \text{via } B \rightarrow bYd \\ \implies & aXbYd \quad \text{via } B \rightarrow bYd \\ \implies & abXYd \quad \text{via } Xb \rightarrow bX \\ \implies & abYcd \quad \text{via } XY \rightarrow Yc \\ \implies & abcd \quad \text{via } Y \rightarrow \lambda \end{array}$$

Problem #2

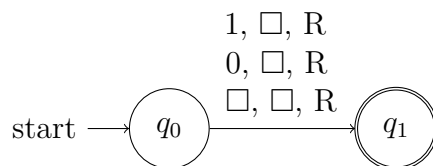
	S	
$\Rightarrow$	TbC	via $S \rightarrow TbC$
$\Rightarrow$	gC	via $Tb \rightarrow g$
$\Rightarrow$	Sg	via $gC \rightarrow Sg$
$\Rightarrow$	$TbCg$	via $S \rightarrow TbC$
$\Rightarrow$	gCg	via $Tb \rightarrow g$
$\Rightarrow$	Sgg	via $gC \rightarrow Sg$
$\Rightarrow$	$TbCgg$	via $S \rightarrow TbC$
$\Rightarrow$	$gCgg$	via $Tb \rightarrow g$
$\Rightarrow$	egg	via $gC \rightarrow e$

Problem #3

The language is derived by the grammar is $L = \{aad^nb^k \mid 0 \leq n \leq k, n+k = 2c\}$, for some arbitrary constant c (i.e. $n+k$ is even).

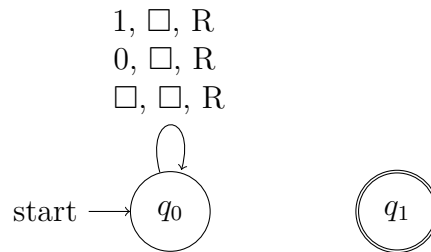
Problem #4

The following always halts, regardless of the input from $\Sigma = \{0, 1\}$.



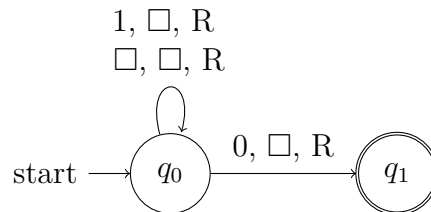
Problem #5

The following never halts, regardless of the input from $\Sigma = \{0, 1\}$. Note it also never halts even if the tape is blank (λ).



Problem #6

The following halts for some, but not all, input from $\Sigma = \{0, 1\}$. In this case, the only input that would cause the machine to halt would be 0.



Problem #7

No, we will prove so as follows.

Theorem 1. *Suppose A to be an arbitrary program — that is, $A \in$ all programs. Then, $\forall A \in$ all program, there **does not** exists a program P such that P can determine if A will halt **regardless of input**.*

Proof. Suppose not. That is, suppose $\exists P \in$ all programs such that P can determine if $\forall A \in$ all programs, A will halt. Suppose the following to be such program P , with an additional source code at the end:

```

bool willHalt(program P) { ... }

int recursiveNightmare() {
    if (willHalt(P)) {
        return recursiveNightmare();
    }

    return 42;
}
  
```

If we were to run this program through P , that is $P(P)$, we have the following cases:

Case 1: The program P loops forever. Assuming the program to be correct, this will return false; hence program P halts on input P , which is a contradiction (by our **recursive nightmare**).

Case 2: The program halts on input P . Assuming the program to be correct, this will return true; hence program P runs indefinitely on input P ; this is a contradiction.

We see this to be a paradox; P only halts when `willHalt` will not halt, and runs indefinitely when it does halt. Either way, we see this to be a contradiction. Therefore, there **does not** exist a program P such that P can determine if A will halt **regardless of input**.

□